SECTION 1 — Time---30 minutes, 45 Questions

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

(A) ● (C) (D) (E)

1. WOE : (A) isolation (B) indecision (C) joy (D) perplexity (E) preference

2. DEPRECIATE : (A) prove (B) attack (C) dawdle (D) locate (E) enhance

3. LINGER : (A) enroll (B) create (C) depart (D) dishearten (E) accept

4. OBJECTIVE : (A) biased (B) sincere (C) tolerable (D) subdued (E) dark

5. FLAUNT : (A) prepare (B) pander (C) remark (D) request (E) cover

6. FLAMBOYANCE : (A) preference (B) dullness (C) shame (D) vitality (E) formality

7. ACCELERATE : (A) garble (B) fret (C) fluctuate (D) withdraw (E) slow

8. LUSH : (A) barren (B) insecure (C) harmful (D) extreme (E) astronomical

9. RECLUSIVE : (A) hopeful (B) supportive (C) unfaithful (D) sociable (E) aggressive

10. MEANDER : (A) travel directly (B) insinuate (C) dance gracefully (D) unblock (E) immerse

11. FLOUT : (A) make irrelevant (B) scatter (C) honor (D) overlook (E) remain innocent

12. EMPIRICAL : (A) farsighted (B) theoretical (C) completed (D) radical (E) farcical

13. SULLY : (A) punctuate (B) suffer (C) purify (D) acquiesce (E) facilitate

14. EVANESCENT : (A) permanent (B) silent (C) adroit (D) habitual (E) restless

15. PARSIMONIOUS : (A) munificent (B) relaxed (C) impartial (D) corpulent (E) courageous

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

● (B) (C) (D) (E)

16. Since the island soil has been barren for so many years, the natives must now ---- much of their food.

(A) deliver (B) import (C) produce (D) develop (E) utilize

17. Although a few critics praised the movie, the tone of most reviews was one of ----.

(A) authority (B) confusion (C) ridicule (D) disbelief (E) caution

18. Since Jenkins neither ---- nor defends either management or the striking workers, both sides admire his journalistic ----.

(A) criticizes..acumen (B) attacks..neutrality (C) confronts..aptitude (D) dismisses..flair (E) promotes..integrity

19. Some anthropologists claim that a few apes have been taught a rudimentary sign language, but skeptics argue that the apes are only ---- their trainers.

(A) imitating (B) mocking (C) instructing (D) acknowledging (E) belaboring

20. It is ironic that the ---- insights of the great thinkers are repeated so often that they have become mere ----.

(A) original..clichés
(B) banal..beliefs
(C) dubious..habits
(D) philosophical..questions
(E) abstract..ideas

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Select the word or set of words that best completes each of the following sentences.

21. The most frustrating periods of any diet are the inevitable ----, when weight loss ---- if not stops.

(A) moods..accelerates (B) feasts..halts
(C) holidays..falters (D) plateaus..slows
(E) meals..tapers

22. Since the author's unflattering references to her friends were so ----, she was surprised that her ---- were recognized.

(A) laudatory..styles
(B) obvious..anecdotes
(C) oblique..allusions
(D) critical..eulogies
(E) apparent..motives

23. Because even the smallest mistakes infuriated him, he reviewed all his papers ---- before submitting them to his teacher.

(A) explicitly (B) perfunctorily (C) honestly
(D) mechanically (E) assiduously

24. The subtleties of this novel are evident not so much in the character ---- as they are in its profoundly ---- plot structure.

(A) psychology..eclectic
(B) development..trite
(C) portrayal..aesthetic
(D) delineation..intricate
(E) illustration..superficial

25. Since many disadvantaged individuals view their situations as ---- as well as intolerable, their attitudes are best described as ----.

(A) squalid..obscure
(B) unpleasant..bellicose
(C) acute..sanguine
(D) immutable..resigned
(E) political..perplexed

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

(A) (B) ● (D) (E)

26. SHIP : OCEAN :: (A) fish : gill
(B) plane : air (C) child : bath
(D) camel : water (E) car : passengers

27. TUNE : PIANO :: (A) eat : food
(B) sharpen : knife (C) record : song
(D) display : painting (E) listen : radio

28. SISTER : SIBLING :: (A) couple : wedding
(B) family : relationship (C) parent : son
(D) subordinate : ruler (E) wife : spouse

29. CENSUS : POPULATION :: (A) catalog : pictures
(B) inventory : supplies (C) detonation : explosion
(D) dictionary : words (E) election : tally

30. TAPESTRY : THREAD :: (A) pizza : pie
(B) mosaic : pieces (C) ruler : divisions
(D) computer : switch (E) car : engine

31. CONSTELLATION : STARS :: (A) earth : moon
(B) center : circle (C) archipelago : islands
(D) rain : water (E) maverick : herd

32. PERSPICACIOUS : INSIGHT ::
(A) zealous : mobility
(B) audacious : hearing
(C) delicious : taste
(D) avaricious : generosity
(E) amiable : friendliness

33. FODDER : HORSE :: (A) gas : automobile
(B) wool : sheep (C) coop : poultry
(D) mother : child (E) doe : deer

34. REFINE : OIL :: (A) winnow : wheat
(B) harness : energy (C) mine : coal
(D) mold : plastic (E) conserve : resource

35. VAGRANT : DOMICILE :: (A) pagan : morals
(B) despot : leadership (C) arsonist : fire
(D) exile : country (E) telephone : ear

GO TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

He walked through the streets of New York wearing a broad-brimmed hat, a dark business suit, a pale green satin shirt, a vest of white brocade, a huge black bow emerging
Line from under his chin; and he carried a staff, not a cane, but an
(5) ebony staff surmounted by a bulb of solid gold. It was as if his huge body were resigned to the conventions of a prosaic civilization and its drab garments, but the oval of his stomach sallied forth, flying the colors of his inner soul.

These things were permitted to him, he thought to
(10) himself, because he was a genius. He was also president of the Architects' Guild of America. Ralston Holcombe did not subscribe to the views of his colleagues in the organization, for he was not a grubbing builder or a businessman. He was, he stated firmly, a man of ideals.

(15) He denounced the deplorable state of American architecture. In any period of time, he declared, architects should build in the spirit of their own time and should not pick designs from the past. We could be true to history only by heeding her law, which demanded that we plant the roots of
(20) our art firmly in the reality of our own life. He decried the stupidity of erecting buildings that were Greek, Gothic, or Romanesque. Let us, he begged, abandon such unprincipled eclecticism. We are modern and must build in the style that belongs to our days. He had found that style--it was
(25) Renaissance.

He stated his reasons clearly. Inasmuch, he pointed out, as nothing of great historical importance had happened in the world since the Renaissance, we should consider ourselves still living in that period, and all the outward forms of our
(30) existence should remain faithful to the examples of the great masters of the sixteenth century.

He had no patience with the few who spoke of a modern architecture in terms quite different from his own; he ignored them; he stated that men who wanted to break with all of the
(35) past were lazy ignoramuses, and that one could not put originality above beauty. His voice trembled reverently on that last word.

He built like a composer improvising under the spur of a mystic guidance. He had sudden inspirations. He would add
(40) an enormous dome to the flat roof of a finished structure, or encrust a long vault with a gold-leaf mosaic. His clients turned pale, stuttered--and paid. His imperial personality carried him to victory in any encounter with a client's thrift; behind him stood the stern, unspoken, overwhelming assertion
(45) that he, above all others, was an Artist.

36. Which of the following titles would be the most appropriate for this passage?

(A) Ralston Holcombe, Architect for Our Times
(B) Ralston Holcombe, Spokesman for the Renaissance
(C) The Deplorable State of American Architecture
(D) A Portrait of Pomposity
(E) Inspiration in Modern Architecture

37. The author implies that Holcombe's torso

(A) is a reflection of his status as an architect
(B) provides a means of expressing his intelligence
(C) is an external symbol of his individuality
(D) overwhelms others despite his desire to conceal it
(E) represents his importance

38. By "eclecticism" (line 23), the author means

(A) designing buildings without regard for principles
(B) creating buildings that employ styles of the past
(C) redefining previous periods of architecture
(D) ignoring historical precedents
(E) respecting the contributions of early artists

39. The author implies that the attitude of Holcombe's clients toward his "inspirations" is usually one of

(A) joy and anticipation
(B) shock and resignation
(C) discomfort and tolerance
(D) surprise and curiosity
(E) disgust and disillusionment

40. Which of the following best describes Holcombe's approach to the world of architecture?

(A) Defiant arrogance
(B) Sympathetic understanding
(C) Intrigued curiosity
(D) Respectful tolerance
(E) Enthusiastic inspiration

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It is easy to lose patience with science today. The questions are pressing: How dangerous is dioxin? What about low-level radiation? When will that monstrous earthquake strike California? And why can't we predict the weather better? But the evidence is often described as "inconclusive," forcing scientists to base their points of view almost as much on intuition as on science.

When historians and philosophers of science listen to this cacophony, some conclude that science may be incapable of solving all these problems any time soon. Many questions seem to defy the scientific method, an approach that works best when it examines straightforward relationships: If variable A is manipulated, what does this do to variable B? Such procedures can, of course, be very difficult in their own ways, but in experimental terms they are clean, elegant.

With the aid of Newton's laws of gravitational attraction, for instance, ground controllers can predict the path of a planetary probe with incredible accuracy. They do this by calculating, one at a time, the gravitational tugs from each of the passing planets until the probe speeds beyond the edge of the solar system. A much more difficult task is to calculate how two or three such tugs work at once. The unknowns can grow until riddles are impossibly knotty. Because of the turbulent and fickle whorls of the earth's atmosphere, for instance, scientists have struggled for centuries to predict the weather with precision.

This spectrum of explanatory power--from simple problems to those impossibly complex--has resulted in nicknames for fields of inquiry. "Hard" sciences, such as astronomy and chemistry, are said to yield precise answers, whereas "soft" ones, such as sociology and economics, admit a great degree of uncertainty.

41. The author's attitude toward those who have lost patience with science can best be described as

(A) angry
(B) patronizing
(C) dismayed
(D) confused
(E) sympathetic

42. The author implies that when confronted with complex questions, scientists base their opinions

(A) on theoretical foundations
(B) more on intuition than on science
(C) on science and intuition, in varying degrees
(D) on observations and past experience
(E) on experimental procedures

43. According to the passage, it can be inferred that the scientific method would work best in which of the following situations?

(A) Predicting public reaction to a set of policy decisions
(B) Identifying the factors which will predict a California earthquake
(C) Predicting the amount of corn that an acre will yield when a particular type of fertilizer is used
(D) Determining the dangers of low-level radiation
(E) Calculating how much a cubic centimeter of water will weigh when cooled under controlled conditions

44. The author suggests that accurately predicting the path of a planetary probe is MORE difficult than

(A) forecasting the weather
(B) determining when an earthquake will occur
(C) predicting economic behavior
(D) explaining why people behave the way they do
(E) determining the gravitational influence of one planet

45. According to the passage, hard science can be distinguished from soft science by which of the following characteristics?

(A) Seeking precise answers to its questions
(B) Identifying important questions that need answers
(C) Making significant contributions to human welfare
(D) Creating debates about unresolved issues
(E) Formulating theories to explain certain phenomena

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 2 — Time---30 minutes, 35 Questions

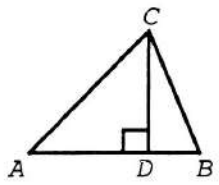
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area = πr^2; Circumference = $2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

= is equal to	≤ is less than or equal to
≠ is unequal to	≥ is greater than or equal to
< is less than	∥ is parallel to
> is greater than	⊥ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\Delta ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

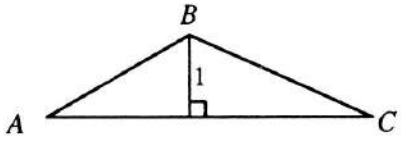
Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. $555{,}555 + (3 \times 10^5) =$

(A) 355,555
(B) 535,555
(C) 558,555
(D) 585,555
(E) 855,555

2. $(7 - 1)(6 - 1)(5 - 1)(4 - 1) - (6 - 1)(5 - 1)(4 - 1) =$

(A) 6 (B) 7 (C) 60 (D) 300 (E) 360



3. If $AC = 4$, what is the area of ΔABC above?

(A) $\frac{1}{2}$ (B) 2 (C) 4 (D) $\sqrt{7}$ (E) 8

4. If $x + y = z$ and $x = y$, then all of the following are true EXCEPT

(A) $2x + 2y = 2z$ (B) $x - y = 0$ (C) $x - z = y - z$
(D) $x = \frac{z}{2}$ (E) $z - y = 2x$

Price of Buttons in Store X	
Color	Price
Black	\$2 per 5 buttons
Blue	\$2 per 6 buttons
Brown	\$3 per 8 buttons
Orange	\$4 per 12 buttons
Red	\$4 per 7 buttons

5. In store X, which button costs the most per individual unit?

(A) Black (B) Blue (C) Brown
(D) Orange (E) Red

6. Which of the following numbers can be written in the form 6K + 1, where K is a positive integer?

(A) 70
(B) 71
(C) 72
(D) 73
(E) 74

7. What is the diameter of a circle with circumference 5?

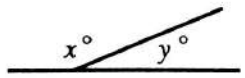
(A) 25π (B) 10π (C) 5π (D) $\frac{5}{\pi}$ (E) $\frac{1}{5\pi}$

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Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

EXAMPLES	Column A	Column B	Answers
E1.	2×6	$2 + 6$	(B) (C) (D) (E)
E2.	 $180 - x$	y	(A) (B) (C) (D) (E)
E3.	$p - q$	$q - p$	(A) (B) (C) (E)

Notes:
1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

	Column A	Column B
8.	$\frac{3}{7}$	$\frac{7}{14}$

$7a > 4b$

9.	a	b

A cube has one red face, one white face, and the remaining faces are blue.

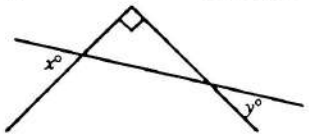
10.	The number of faces that are blue	3

Michael had \$60, and lent one-third of it to Greg, who now has \$50.

11.	Amount of money Greg originally had	Amount of money Michael now has

12.	$\sqrt{3} + \sqrt{4}$	$\sqrt{5}$

	Column A	Column B
13.	$x + y$	90



14.	xy	$x\sqrt{y}$



The novelty clock above has hands which move at the correct speed, but counter-clockwise. The clock tells the correct time every 6 hours (at 6:00 and 12:00).

15.	3 hrs. 15 minutes	Number of hours that have passed since 12:00

$9^n - 8^n = 1^n$

16.	1	n

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

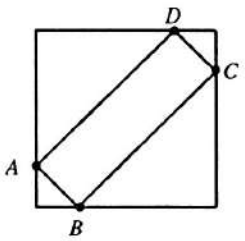
AN E RESPONSE WILL NOT BE SCORED

	Column A	Column B
	A rectangle of area 4 has two sides of length r and s, where r and s are integers.	
17.	r	s

List 1
1
2
3
4
5

List 2
2
3
4
5
6

	Column A	Column B
18.	The product of four different numbers from list 1	The product of four different numbers from list 2



The area of the square is 25. Points A, B, C, and D are on the square.

	Column A	Column B
19.	20	Perimeter of the rectangle $ABCD$

$x = 3x + 1$

	Column A	Column B
20.	$-x$	x

	Column A	Column B
	x, y and z are positive. $x + y + z = 10$ and $x = y$	
21.	x	5
22.	The least even integer greater than -2	0

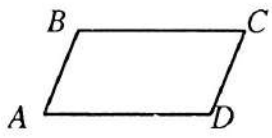
Cable lines cost \$40 per 100 meters.

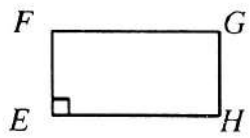
	Column A	Column B
23.	The cost in dollars of m meters of cable line	$0.4m$
24.	The number of distinct prime factors of 60.	The number of distinct prime factors of 30.
25.	x^2	$(x + 1)^2$

$\frac{(a+b)}{c} = 0$

$(a + c) - b = a$

	Column A	Column B
26.	a	$-c$

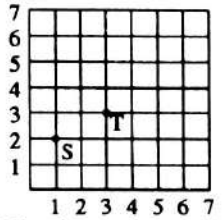




	Column A	Column B
27.	Area of parallelogram $ABCD$ if $AD = 1$ and $CD = w$ and $\angle ADC \neq 90$	Area of rectangle $EFGH$ if $EH = 1$ and $GH = w$

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Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.



28. Which of the following points will lie on the line drawn through points S and T?

(A) (1, 1) (B) (3, 1) (C) (3, 2)
(D) (4, 4) (E) (5, 4)

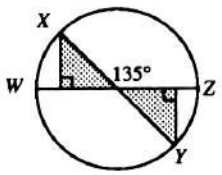
29. The difference between $7\frac{1}{4}$ hours and $6\frac{3}{5}$ hours is how many minutes?

(A) 30 (B) 36 (C) 39 (D) 45 (E) 51

For all integers n not equal to 1, $\langle n \rangle = \frac{n+1}{n-1}$

30. Which of the following has the greatest value?

(A) $\langle 0 \rangle$ (B) $\langle 2 \rangle$ (C) $\langle 3 \rangle$ (D) $\langle 4 \rangle$ (E) $\langle 5 \rangle$



31. If segment WZ and segment XY are diameters of the circle with length 12, what is the area of the shaded triangles?

(A) 36 (B) 33 (C) 30 (D) 18 (E) 12

32. If $w \neq 0$ and $w = 2x = 3y$, what is the value of $w + x$ in terms of y?

(A) $\frac{1}{5}y$ (B) y (C) $\frac{9}{2}y$ (D) $3y$ (E) $6y$

33. If $y = (x + 2)^2 + 1$, what is the minimum value of y?

(A) -2
(B) -1
(C) 0
(D) 1
(E) 5

34. Amy owes Berna \$2, Berna owes Cindy \$10, and Cindy in turn owes Amy some money. How much does Cindy owe Amy if all three debts could be settled by having Berna pay \$1 to Amy and \$7 to Cindy?

(A) \$2 (B) \$3 (C) \$4 (D) \$5 (E) \$6

35. On a map, 1 centimeter represents 6 kilometers. A square on the map with a perimeter of 16 centimeters represents a region with what area?

(A) 64 km^2
(B) 96 km^2
(C) 256 km^2
(D) 576 km^2
(E) 8216 km^2

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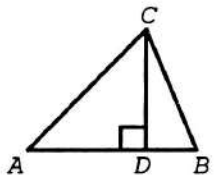
SECTION 5 | Time---30 minutes, 25 Questions | In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

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Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

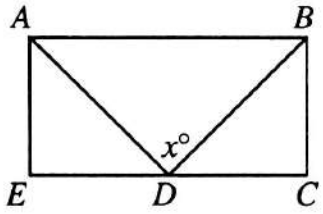
(1) area of $\Delta ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $9b = 81$, then $3 \times 3b =$

(A) 9
(B) 27
(C) 81
(D) 243
(E) 729



2. In the figure above, if $AE = ED = DC = CB$, then $x =$

(A) 60 (B) 75 (C) 90 (D) 120 (E) 135

3. $\frac{0.5 + 0.5 + 0.5 + 0.5}{4} =$

(A) .05 (B) .25 (C) .5 (D) 1 (E) 2.0

4. For any set of numbers S, $M(S)$ is defined as the maximum number of occurrences of any number in the set. For example, if $S = \{1, 2, 2, 2\}$ then $M(S) = 3$. If $S = \{1, 2, 3, 3, 4, 5, 5, 6\}$ then $M(S) =$

(A) 0
(B) 2
(C) 4
(D) 6
(E) 8



Note: Figure not drawn to scale.

5. If $AB > CD$, which of the following must be true?

I. $AB > BC$
II. $AC > BD$
III. $AC > CD$

(A) I only (B) II only (C) III only
(D) II and III only (E) I, II, and III

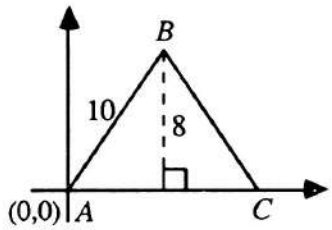
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6. If 3 more than x is 2 more than y, what is x in terms of y?

(A) $y - 5$
(B) $y - 1$
(C) $y + 1$
(D) $y + 5$
(E) $y + 6$

7. If 8 and 12 each divide K without a remainder, what is the value of K?

(A) 16 (B) 24 (C) 48 (D) 96
(E) It cannot be determined from the information given.



8. Side AB of the triangle ABC above must contain which of the following points?

(A) (3, 2) (B) (3, 5) (C) (4, 6)
(D) (4, 10) (E) (6, 8)

9. If January 30th occurs on a Monday, on what day did January 1st occur?

(A) Sunday
(B) Monday
(C) Tuesday
(D) Friday
(E) Saturday

10. Ricky bought grapes at a price of \$1.199 per pound. Of the following, which is closest to the number of pounds Ricky bought if he paid \$36 for the grapes?

(A) 31 (B) 30 (C) 29 (D) 28 (E) 18

11. Which of the following is closest to 4?

(A) 3.4
(B) 4.04
(C) 4.404
(D) 3.94
(E) 4.004

12. To the nearest thousand, what is the number of hours in a decade?

(A) 8,000
(B) 87,000
(C) 88,000
(D) 89,000
(E) 90,000

13. The Daily Newspaper has a particular format it likes to follow. There are six articles on the pages with even numbers, and seven articles on the pages with odd numbers. How many articles are there in today's paper if the paper has fifty-one pages beginning with page number one?

(A) 325 (B) 331 (C) 332 (D) 338 (E) 339

14. The product of (1 + 2), (2 + 3), and (3 + 4) is equal to half the sum of 20 and

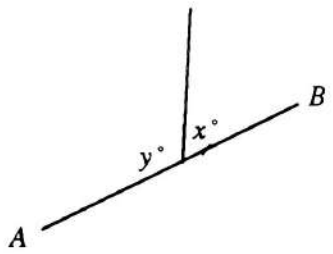
(A) 3
(B) 85
(C) 105
(D) 190
(E) 1210

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15. Which of the following must be true?

I. The sum of two consecutive integers is odd.
II. The sum of three consecutive integers is even.
III. The sum of three consecutive integers is a multiple of 3.

(A) I only (B) II only (C) I and II only
(D) I and III only (E) I, II, and III



Note: Figure not drawn to scale.

16. In the figure above, if AB is a line segment and $3x = y$, what is the value of y?

(A) 45 (B) 60 (C) 90 (D) 120 (E) 135

17. What are all values of x such that $x^2 - 3x - 4$ is negative?

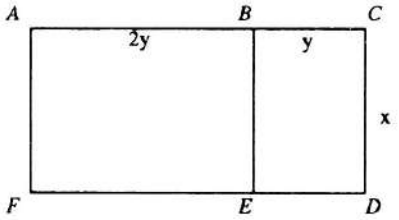
(A) $x < -1$ or $x > 4$
(B) $x < -4$ or $x > 4$
(C) $1 < x < 4$
(D) $-4 < x < 1$
(E) $-1 < x < 4$

18. At a certain school, 30 students study French, 40 study Spanish, 5 study both, and 10 study neither. How many students are at this school?

(A) 70 (B) 75 (C) 80 (D) 85 (E) 90

19. If c is positive, what percent of $3c$ is 9?

(A) $\frac{c}{100}\%$ (B) $\frac{c}{3}\%$ (C) $\frac{9}{c}\%$
(D) 3% (E) $\frac{300}{c}\%$



20. In the figure above, the area of rectangle $ACDF$ is how many times as great as the area of rectangle $BCDE$?

(A) 18 (B) 6 (C) 3 (D) 2
(E) It cannot be determined from the information given.

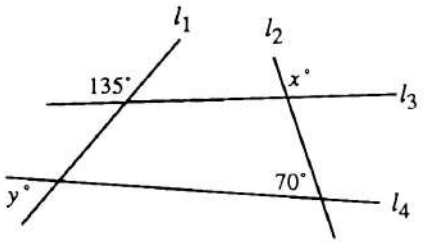
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21. The total number of integers between 1 and 200 inclusive that equal the cube of an integer is

(A) one (B) two (C) three
(D) four (E) five

22. If the price of dress A is three times the average price of 10 other dresses, what fraction of the total price of the 11 dresses is the price of dress A?

(A) $\frac{2}{13}$ (B) $\frac{3}{11}$ (C) $\frac{3}{13}$ (D) $\frac{3}{10}$ (E) $\frac{3}{4}$



Note: Figure not drawn to scale

23. If four lines intersect as shown in the figure above, $x + y =$

(A) 65 (B) 110 (C) 155 (D) 205
(E) It cannot be determined from the information given.

24. If $\frac{1}{x} + \frac{1}{y} = 6$ and $xy = 8$, what is the average (arithmetic mean) of x and y?

(A) -7 (B) 12 (C) 14 (D) 24 (E) 48

25. If $4 < a < 7 < b < 9$, then which of the following best defines $\frac{a}{b}$?

(A) $\frac{4}{9} < \frac{a}{b} < 1$

(B) $\frac{4}{9} < \frac{a}{b} < \frac{7}{9}$

(C) $\frac{4}{7} < \frac{a}{b} < \frac{7}{9}$

(D) $\frac{4}{7} < \frac{a}{b} < 1$

(E) $\frac{4}{7} < \frac{a}{b} < \frac{9}{7}$

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. STOP

SECTION 6 | Time---30 minutes 40 Questions | For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

1. NEGLECT : (A) innocence (B) attention (C) expression (D) irrelevance (E) distraction

2. HUMID : (A) fresh (B) mild (C) cool (D) dry (E) feeble

3. DEFY : (A) submit (B) request (C) compete (D) cleverly deceive (E) allow disagreement

4. DEBASE: (A) uplift (B) assume (C) obey (D) decry (E) reject

5. RESERVE : (A) opposition (B) support (C) notice (D) injury (E) recklessness

6. GARISH: (A) adaptable (B) sedate (C) explicable (D) generous (E) nonchalant

7. IMPOSING: (A) uproarious (B) narrow (C) insignificant (D) random (E) secretive

8. PREDISPOSITION: (A) retention (B) delay (C) aversion (D) dissension (E) openmindedness

9. LATENT : (A) upright (B) straight (C) obvious (D) faithful (E) borne

10. VACILLATE:
(A) hold fast
(B) learn well
(C) inflict
(D) upset
(E) peruse

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Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

●(B)(C)(D)(E)

11. If it is true that morality cannot exist without religion, then does not the erosion of religion herald the ---- of morality?

(A) regulation (B) basis (C) belief (D) collapse (E) value

12. I don't believe that a sense of justice is ----, but I have been astonished to see how quickly it can be taught.

(A) practical (B) innate (C) intellectual (D) manifest (E) universal

13. The former general led his civilian life as he had his military life, with simplicity and ---- dignity.

(A) benevolent (B) informal (C) austere (D) aggressive (E) succinct

14. Only when one actually vists the ancient ruins of marvelous bygone civilizations does one truly appreciate the sad ---- of human greatness.

(A) scope (B) magnitude (C) artistry (D) transience (E) innovation

15. Most students found Dr. Schwartz's lecture on art excessively detailed and academic; some thought this display of ---- exasperating.

(A) rationalism (B) erudition (C) logic (D) pedantry (E) literalism

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

(A)(B)●(D)(E)

16. MUSICIAN : ORCHESTRA :: (A) coach : football (B) lawyer : jury (C) poet : writing (D) actor : troupe (E) pediatrician : doctor

17. BREEZE : HURRICANE :: (A) water : pebble (B) gulf : coast (C) eye : cyclone (D) sun : cloud (E) hill : mountain

18. IMMORTAL : DEATH :: (A) anonymous : fame (B) hopeless : situation (C) vital : life (D) indisputable : agreement (E) daily : year

19. DUEL : FIGHT :: (A) debate : argument (B) contest : referee (C) recess : intermission (D) match : tournament (E) act : play

20. LUBRICANT : FRICTION :: (A) motor : electricity (B) speed : drag (C) insulation : heat (D) adhesive : connection (E) muffler : noise

21. PARODY : IMITATION :: (A) stanza : verse (B) limerick : poem (C) novel : book (D) portrait : painting (E) riddle : puzzle

22. COMET : TAIL :: (A) traffic : lane (B) missile : trajectory (C) vessel : wake (D) engine : fuel (E) wave : crest

23. NEOLOGISM : LANGUAGE ::
(A) rhetoric : oratory
(B) syllogism : grammar
(C) innovation : technology
(D) iconography : art
(E) epistemology : philosophy

24. ADDENDUM : BOOK :: (A) signature : letter (B) vote : constitution (C) codicil : will (D) heading : folder (E) stipulation : contract

25. AXIOM : MATHEMATICS :: (A) tenet : creed (B) equation : physics (C) analysis : reasoning (D) cadence : music (E) compound : chemistry

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Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

I was brought up to believe that sports would be the salvation of the black race in this country. As I grew to manhood, I was told repeatedly that sports represented the one area of American life in which a black man could proceed according to his own merit. For proof, I had only to look at the careers of exceptional individuals like Jesse Owens, Joe Louis, Jackie Robinson, and Willie Mays, with special attention to Paul Robeson at Rutgers and Jerome Holland at Cornell. I accepted the evidence, and when Holland was awarded the position of United States ambassador to Sweden, what had been up to that time merely a theory became for me an established fact. I often assured my friends that "for the black man, sports are the highroad to upward escalation, the proven way he will escape from the ghetto."

There was, of course, that distinctly awkward afternoon when a renowned sociologist rebuked me in an extremely serious manner: "All that you are saying is that if a young black possesses superhuman capabilities, if he can run like Jesse Owens, or play football like Jerome Holland, he can win for himself the normal decencies that the ordinary white man in this country takes for granted. We will have no social justice in this country until the day when an ordinary black, with no more than a ninth-grade education behind him and no outstanding skills, has full equality here on every level."

For a disturbing moment, I felt that I had caught a glimpse of what this radical scholar was proposing, but in my innocence I still dismissed both him and his argument. After all, the man was a known trouble-maker and might even have been a Communist. I reminded myself that the facts of the situation were irrefutable: blacks in the United States were finding a place for themselves in sports, and this would be their ultimate salvation. It was only years later that I discovered the truth of his words.

26. The author's attitude toward Jesse Owens, Joe Louis, and Jackie Robinson was one of

(A) distaste
(B) admiration
(C) outrage
(D) envy
(E) ambivalence

27. The author considered the accomplishments of Jerome Holland to be

(A) a sign that blacks were discriminated against
(B) evidence that blacks could accomplish a great deal through sports
(C) troubling in that Holland had to demonstrate superhuman capabilities
(D) unbelievable and unlikely to be repeated
(E) proof that his sociologist friend was right

28. When he writes that the sociologist "might even have been a Communist," the author is being

(A) speculative
(B) ironic
(C) laudatory
(D) political
(E) vague

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Charles Dickens, like untold millions of children all over the world throughout the ages, was enchanted by fairy tales. He acknowledged the deep formative impact that the wondrous figures and events of fairy tales had on his childhood and on his creative genius. Dickens understood that the imagery of fairy tales helps children better than anything else in their most difficult, yet most important and satisfying task: achieving a more mature consciousness to tame the chaotic pressures of their unconscious.

Fairy tales, unlike any other form of literature, direct children to discover their identity and calling. These stories hint that a good, rewarding life is within one's reach despite adversity. They promise that if one dares to engage in this fearsome and taxing search, benevolent powers will come to one's aid, and that one will succeed. But fairy tales also warn that those who are too timid or narrow-minded to risk themselves must settle for humdrum existence.

In the past, those who loved fairy tales were often subjected to the scorn of pedants. But today many of our children are far more grieviously bereaved; they are deprived of the chance to know fairy stories at all. Most children now meet fairy tales, if they encounter them at all, only in prettified versions that subdue their meaning and rob them of all deeper significance. One can see such versions in films and on television, where fairy tales are turned into empty-minded entertainment.

Throughout history, the intellectual life of children largely depended on mythical and religious stories, and on fairy tales. This traditional literature fed children's imagination and stimulated their fantasies. At the same time, these stories were a major agent of the child's socialization. Children could learn social ideals from the material of myths, while fairy tales provided patterns of behavior modeled on these ideals. These were the images of the unconquered heroes Achilles and Odysseus, and of Hercules, whose life history showed that it is not beneath the dignity of the strongest man to clean the filthiest stable.

29. The author cites Charles Dickens in order to

(A) call attention to a person who began by writing fairy tales
(B) demonstrate how fairy tale themes became incorporated into his work
(C) support the author's scorn for those who adapt fairy tales for film and television
(D) prove the formative influence of fairy tales on writers
(E) provide an example of one who understands the importance of fairy tales to children

30. The author identifies all of the following as benefits children derive from fairy tales EXCEPT

(A) expressing complex issues in language children can understand
(B) serving as introductions to complex but rewarding lifestyles
(C) teaching children the wisdom of facing risks to achieve more fulfilling lives
(D) helping children make sense of their inner conflicts
(E) improving a child's imagination through models of vivid imagery

31. According to the passage, today's children seldom experience fairy tales in the original because these literary forms are

(A) subject to the scorn of child development specialists
(B) no longer available
(C) not understandable to children
(D) diluted to be entertaining
(E) not relevant to the concerns of contemporary childhood

32. Which of the following best expresses the main idea of the passage?

(A) Many of today's children lack the important socializing influences of myths and fairy tales.
(B) Fairy tales provide models for human feelings and are therefore essential to good creative writing.
(C) Families who expose children to the original fairy tales go against the advice of child development specialists.
(D) The best way to tame a child's unconscious being is to allow him to experience fairy tales.
(E) The diminishing importance of the role of fairy tales in child development explains today's problems.

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The Supreme Court's recent decision allowing regional interstate banks has razed one hedge in America's banking maze, although many others still remain. Although the ruling does not apply to very large money-center banks, it is a move in a liberalizing direction that could at last push Congress into framing a sensible legal and regulatory system that allows banks to plan their future beyond the next court case.

The restrictive laws the courts are interpreting are mainly a legacy of the bank failures of the 1930s. The current high rate of bank failures--higher than at any time since the great depression--has made legislators afraid to remove the restrictions. While their legislative timidity is understandable, it is also mistaken. One reason so many American banks are getting into trouble is precisely that the old restrictions make it hard for them to build a domestic base large and strong enough to support their activities in today's telecommunicating, round-the-clock, around-the-world financial markets.

In trying to escape from this straitjacket, banks are taking enormous, and what should be unnecessary, risks. For example, would a large bank be buying small, failed savings banks at inflated prices if federal laws and states regulations permitted that bank to expand instead through the acquisition of financially healthy banks in the region? Of course not.

The solution is clear: American banks will be sounder when they are not geographically limited. The House of Representatives' banking committee has shown part of the way forward by recommending common-sensical, though limited, legislation for a five-year transition to nationwide banking. This would give regional banks time to group together to form counterweights to the big money-center banks. Without this breathing space the big money-center banks might soon sprawl across the country to become oligopolies. But any such legislation should be regarded as only a way station on the road towards a complete overhaul of American's creaking banking legislation.

33. The author's attitude towards the current banking laws is best described as one of

(A) concerned dissatisfaction
(B) tolerant disapproval
(C) uncaring indifference
(D) mild endorsement
(E) great admiration

34. Which of the following best describes why the restrictive banking laws of the 1930's are still on the books?

(A) The bank failures of the 1930's were caused by restrictive courts.
(B) Banking has not changed in the past fifty years.
(C) The Supreme Court rulings have adequately updated the existing laws.
(D) The banking system is too restrictive, but no alternatives have been suggested.
(E) Legislators apparently believe that banking problems similar to those of the depression still exist today.

35. The author argues that the change to a nationwide banking system should be

(A) gradual, so that regional banks have a chance to compete with larger banks
(B) postponed, until the consequences can be evaluated
(C) immediate, because we cannot afford any more bank failures
(D) reconsidered, in light of recent Supreme Court decisions
(E) accelerated, to overcome legislative fear

36. Which of the following best expresses the main idea of the passage?

(A) The current banking laws must be interpreted by the Supreme Court to be useful to today's banks.
(B) Although there are currently many bank failures, the nature of banking has not really changed that much.
(C) Money-center banks currently have too much power as compared with the regional banks.
(D) Because current laws are not responsive to contemporary banking needs, banks have been forced to take needless and dangerous risks.
(E) As a result of the many Supreme Court rulings liberalizing today's banking regulations, bank failures continue to occur.

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Schooling of fish is one of the most familiar forms of animal social behavior, but until recently it was little understood, possibly because of the impossibility of observing minute changes in the position and the velocity of fish in a school under natural conditions. The fact that over the eons a great many species of fish have developed the abilty to congregate in schools suggests that the behavior offers a considerable evolutionary survival advantage over those species that do not form schools. How the school is formed and maintained, however, is only beginning to be understood.

One of the most persistent misconceptions about fish schools is that they have a regular geometric form. Our work shows that the structure is a loose and probabilistic one, and that it results from each fish's applying a few simple rules. The first rule is that for each species there is a characteristic minimum approach distance within which neighbors do not come. The distance is usually about three-tenths of a body length. The minimum approach distance is not the distance that is generally maintained within the fish in a school, however. In each species there is a typical preferred distance to the nearest neighbor, which is usually about one body length. The spatial relations among the fish in a school change constantly as the fish adjust their speed and direction. For this reason the distance to the nearest neighbor is not uniform, even for a single fish. The preferred distance is a statistical abstraction, found by averaging the actual distances over a long period.

Fish of a particular species also tend to keep their nearest neighbor at a particular angle with respect to their body axis. Like the preferred distance, the preferred angle is a statistical quantity. At any given moment only a few fish may have their nearest neighbors at a preferred angle, but over the long period the preferred angle dominates. One useful measure of the degree of a school is the average ratio between the distance to the nearest neighbor and the distance to the second nearest neighbor. The closer the ratio to one, the more uniform the structure. In a cubic lattice the ratio is exactly one. The ratio varies considerably among species. For herring it is about 1.1, for pollack 1.3, and for cod 1.5. A ratio of 1.6 would be observed if the fish in the school took positions at random.

37. The author's observation that schooling is an evolutionary development suggests that

(A) schooling of fish is a specific learned behavior
(B) fish that do not swim in schools will become extinct
(C) fish that swim in schools spend all their time in formation
(D) some species of fish that swim in schools did not always do so
(E) the location of a species of fish determines whether it forms schools

38. By "probabilistic" (line 13) the author means that

(A) his conclusions are uncertain
(B) the structure of a school forms somewhat according to chance
(C) there are statistical methods for determining how a school will form
(D) even species that usually swim in schools occasionally break formation
(E) there are geometric rules for predicting a school's structure

39. It can be inferred that the author is now able to study schooling in fish because

(A) the author employed a new method for observing and measuring small changes in school formations
(B) the author proved that evolution is responsible for the longer periods of time
(C) new technology allows underwater study for longer periods of time
(D) new developments in statistical theory allow him to predict the formation of schools of fish
(E) the author has successfully described the structure of schools of several different species of fish.

40. All of the following are behavioral rules used by fish swimming in the school EXCEPT the

(A) characteristic minimum approach distance
(B) typically preferred distance to the nearest neighbor
(C) depth at which the school swims
(D) speed and direction of movement of the school
(E) preferred angle of body orientation

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. STOP

TEST ANSWERS

Section 1		Section 2		Section 5		Section 6	
1. C	26. B	1. E	26. C	1. C		1. B	26. B
2. E	27. B	2. D	27. B	2. C		2. D	27. B
3. C	28. E	3. B	28. E	3. C		3. A	28. B
4. A	29. B	4. E	29. C	4. B		4. A	29. E
5. E	30. B	5. E	30. B	5. D		5. E	30. A
6. B	31. C	6. D	31. D	6. B		6. B	31. D
7. E	32. E	7. D	32. C	7. E		7. C	32. A
8. A	33. A	8. B	33. D	8. E		8. E	33. A
9. D	34. A	9. D	34. B	9. A		9. C	34. E
10. A	35. D	10. A	35. D	10. B		10. A	35. A
11. C	36. D	11. B		11. E		11. D	36. D
12. B	37. C	12. A		12. C		12. B	37. D
13. C	38. B	13. C		13. C		13. C	38. B
14. A	39. B	14. D		14. D		14. D	39. A
15. A	40. A	15. B		15. D		15. D	40. C
16. B	41. E	16. C		16. E		16. D	
17. C	42. C	17. D		17. E		17. E	
18. B	43. E	18. D		18. B		18. A	
19. A	44. E	19. A		19. E		19. A	
20. A	45. A	20. A		20. C		20. E	
21. D		21. B		21. E		21. B	
22. C		22. C		22. C		22. C	
23. E		23. C		23. C		23. C	
24. D		24. C		24. D		24. C	
25. D		25. D		25. A		25. A	

TEST SCORE KEY

	Reported Score			Reported Score			Reported Score	
Raw Score	SAT-Verbal	SAT-Math	Raw Score	SAT-Verbal	SAT-Math	Raw Score	SAT-Verbal	SAT-Math
85	800		50	510	680	15	280	360
84	780		59	500	670	14	280	350
83	770		48	500	660	13	270	340
82	760		47	490	650	12	260	340
81	750		46	490	640	11	260	330
80	740		45	480	640	10	250	320
79	730		44	470	630	9	240	310
78	720		43	470	620	8	230	310
77	710		42	460	610	7	230	300
76	700		41	450	600	6	220	290
75	690		40	450	590	5	210	280
74	680		39	440	580	4	210	280
73	680		38	440	570	3	200	270
72	670		37	430	560	2	200	260
71	660		36	420	550	1	200	250
70	650		35	420	540	0	200	250
69	650		34	410	530	−1	200	240
68	640		33	410	530	−2	200	230
67	630		32	400	520	−3	200	230
66	620		31	390	510	−4	200	220
65	610		30	390	500	−5	200	210
64	610		29	380	490	−6	200	200
63	600		28	370	480	or		
62	590		27	370	470	below		
61	580		26	360	460			
60	580	800	25	350	450			
59	570	780	24	350	440			
58	560	760	23	340	430			
57	560	750	22	330	420			
56	550	740	21	330	410			
55	540	730	20	320	410			
54	540	720	19	310	400			
53	530	710	18	310	390			
52	520	700	17	300	380			
51	520	690	16	290	370			